
Feed Pump P-450 — Condition Monitoring & Maintenance Guide (excerpt)

Supplied grounding document for capstone groups (e.g. a predictive-maintenance agent) that don't have their own. P-450 is the beer-feed pump to distillation column D-201.

1. Normal operating envelope

- Bearing vibration (velocity RMS): ≤ 4.5 mm/s normal; **alarm at 7.1 mm/s; trip at 11.0 mm/s** (ISO 10816 zones).
- Bearing temperature: ≤ 80 °C normal; alarm at 90 °C.
- Suction pressure: ≥ 0.8 bar (below this risks cavitation).
- Seal-pot level: within the marked band; a falling level indicates seal leakage.
- Delivers beer feed to D-201 at ~ 12 m³/h (see SOP-D201-014 §3).

2. Troubleshooting — rising vibration

1. **Cavitation** — if suction pressure < 0.8 bar with rising vibration and noise, check the strainer for blockage; do not throttle suction further.
2. **Bearing wear** — steadily rising vibration + temperature over days points to bearing degradation; schedule replacement at the next window before the 7.1 mm/s alarm.
3. **Misalignment / imbalance** — a sudden step change after maintenance; recheck alignment.
4. **Impeller fouling / damage** — falling flow at constant speed with vibration.

3. Escalation & change control

- Any decision to keep running above the 7.1 mm/s alarm, or to shut down for maintenance, requires **maintenance-engineer sign-off**.
- The assistant must not command a shutdown or a speed change directly — it recommends, cites this guide, and flags for human approval.